Chia Sengzhi
CID 00802926

Probability & Stochastic Processes Coursework 2016-2017

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Random Variables

1. a) $X = N(5, 2)$

$Y = 2X + 4$

$\Rightarrow Y = N(14, 8)$

$= N(2\times5+4, (2)^2 \times 2)$

$E(Y) = E(2X+4) = 2E(X) + 4 = 14$

$Var(Y) = E(Y^2) - E(Y)^2 = E(4X^2 + 8X + 16) - (2E(X)+4)^2 = 4E(X^2) + 8E(X) + 16 - 4E(X^2) - 8E(X) - 16$

$= 4(E(X^2) - E(X)^2)$

$= 4(2)$

Multiplying a random gaussian variable by a scalar scales the mean by the same scalar & the variance is scaled by the square of the scalar.

Adding a constant only increases the mean by the constant but does not change the variance

$\therefore$ mean = 14, standard deviation = $\sqrt{8}$, distribution = $N(14, 8)$ ✓ ✓ ✓ -2

b) $x = U(0,1) \Rightarrow f_x(x) = 1$

$y = -\ln x \Rightarrow \frac{dx}{dy} = -\frac{1}{y}$

$f_y(y) = \sum_i \left|\frac{dx}{dy}\right| f_x(x)$

$= \left|\frac{dx}{dy}\right| f_x(x)$ $-\ln x$ is monotonic

$= \frac{1}{y}$ ✓ -4

c) i) $z = x + y \Rightarrow f_z(z) = \int_{x,y} f(x, z-x)\,dx = \int_0^z f(x) f_y(z-x)\,dx$ $(x, y > 0) \Rightarrow (z > 0)$

$= \int_0^z e^{-x} e^{-(z-x)}\,dx = \int_0^z e^{-z}\,dx = xe^{-z}\Big|_0^z = ze^{-z}$ ✓

ii) $z = \min(x, y) \Rightarrow f_z(z) = f_x(z) P(y \geq z) + f_y(z) P(x \geq z)$

$= e^{-z}\int_z^\infty e^{-y}\,dy + e^{-z}\int_z^\infty e^{-x}\,dx$

$= 2e^{-z}\left(-e^{-x}\Big|_z^\infty\right)$

$= 2e^{-z}\left(e^{-z}\right) = 2e^{-2z}$ ✓

iii) $z = \max(x, y) \Rightarrow f_z(z) = f_x(z) P(y < z) + f_y(z) P(x < z)$

$= e^{-z}\int_0^z e^{-y}\,dz + e^{-z}\int_0^z e^{-x}\,dx$

$= 2e^{-z}\left(-e^{-x}\Big|_0^z\right)$

$= 2e^{-z}\left(1 - e^{-z}\right)$ ✓

Estimation and sequences of random variables

2. a) likelihood of $x_i = 3.1, 3.5, 3.9$ given $c = \prod_i c^4 x_i^3 e^{-cx_i}$

$$= c^{12} x_1^3 x_2^3 x_3^3 e^{-c(x_1+x_2+x_3)}$$

Maximize log-likelihood $= \ln c^{12} x_1^3 x_2^3 x_3^3 e^{-c(x_1+x_2+x_3)}$

$$= 12 \ln c + 3(\ln x_1 + \ln x_2 + \ln x_3) - c(x_1+x_2+x_3)$$

Take partial derivative with respect to $c \Rightarrow \frac{12}{c} - (x_1+x_2+x_3) = 0$

& set result to 0 $\Rightarrow c = \frac{12}{x_1+x_2+x_3} = \frac{12}{3.1+3.5+3.9}$

≈ 1.22 #4 ✓

b) $Y = \max(x_1, x_2 \ldots x_n)$, $x_i \in (0,1)$

$\Rightarrow F_Y(Y) = F_x(Y)\left[\int_0^y f_x(x)\,dx\right]^{n-1}$

However $F_x(Y) = \int_0^y f_x(x)\,dx = \int_0^y dx = y$ ✓

$\therefore F_Y(Y) = y^n$ // $\quad y \in (0,1)$

c) $E(\hat{\varepsilon}^2) = E\left(\left[s(t-T/2) - as(t) - bs(t-T)\right]^H \left[s(t-T/2) - as(t) - bs(t-T)\right]\right)$

$= E(s(t-T/2)^H s(t-T/2) + a^2 s(t)^H s(t) + b^2 s(t-T)^H s(t-T) - as(t-T/2)^H s(t)$ ~~$as(t)^H s(t-T/2) + ab s(t)^H s(t-T) +$~~

$ab s(t-T)^H s(t) - bs(t-T/2)^H s(t-T) - bs(t-T)^H s(t-T/2)$)

$= R(0) + a^2 R(0) + b^2 R(0) - 2aR(T/2) + 2abR(T) - 2bR(T/2)$ (Assuming s is WSS)

$= R(0)(1+a^2+b^2) - R(T/2)(2a+2b) + 2abR(T)$

$\frac{\partial}{\partial a} \text{Var}\,\varepsilon = 2aR(0) - 2R(T/2) + 2bR(T) = 0$

$\Rightarrow a = \frac{R(T/2) - bR(T)}{R(0)}$ — ①

$\frac{\partial}{\partial b} \text{Var}\,\varepsilon = 2bR(0) - 2R(T/2) + 2aR(T) = 0$

$\Rightarrow b = \frac{R(T/2) - aR(T)}{R(0)}$ — ②

① into ② $a = \frac{R(T/2) - R(T)\left(\frac{R(T/2) - aR(T)}{R(0)}\right)}{R(0)} \Rightarrow aR(0)^2 = R(T/2)R(0) - R(T)R(T/2) + aR(T)^2$

$\Rightarrow a = \frac{R(T/2)R(0) - R(T)R(T/2)}{-R(T)^2 + R(0)^2}$

$\Rightarrow b = \frac{R(T/2) - R(T)\left(\frac{R(T/2) - bR(T)}{R(0)}\right)}{R(0)} \Rightarrow bR(0)^2 = R(T/2)R(0) - R(T)R(T/2) + bR(T)^2$

$\Rightarrow b = \frac{R(T/2)R(0) - R(T)R(T/2)}{R(0)^2 - R(T)^2}$

$a = b = \left(R(T/2)R(0) - R(T)R(T/2)\right)\left(R(0)^2 - R(T)^2\right)^{-1} = \text{II}\left(e^{-\frac{1}{2}} - e^{-\frac{3}{2}}\right)\left(1 - e^{-2}\right)^{-1} = \text{II}\,(0.4434)$ ✓

$\therefore E(\hat{\varepsilon}^2) = R(0)(1+a^2+b^2) - R(T/2)(2a+2b) + 2abR(T)$

$= \text{II}(1 + 2a^2) - \text{II}e^{-\frac{1}{2}}(4a) + 2a^2\text{II}e^{-1}$ $\quad (R(0) = \text{II})$

$= \text{II}\,(0.4621)$ ✓

Random Processes

1. a) i) Fair coin $\Rightarrow$ 0.5 heads, 0.5 tails

$\therefore E[x(t)] = \frac{\sin \pi t}{2} + t$ ✓

ii) At $t = 0.25$, $x(0.25) = \sin\frac{\pi}{4}$ with probability 0.5 & $x(0.25) = 0.5$ with probability 0.5

$F(x, 0.5) = ?$ (-1)

At $t = 0.5$, $x(0.5) = \sin\frac{\pi}{2}$ with probability 0.5 & $x(0.5) = 1$ with probability 0.5

$F(x, 0.5) = ?$ (-1)

$\Rightarrow x(0.5) = 1$ with probability 1

At $t = 1$, $x(1) = \sin\pi = 0$ with probability 0.5 & $x(1) = \lambda$ with probability 0.5

$F(x, 1) = ?$ (-1)

b) i) $P(|x(t+\tau) - x(t)| \geq a) = P([x(t+\tau) - x(t)]^2 \geq a^2)$

By markov inequality, $P(|x(t+\tau) - x(t)| \geq a) \leq \frac{E[x(t+\tau)^2 + x(t)^2 - 2x(t)x(t+\tau)]}{a^2}$

If $x(t)$ is WSS, $P(|x(t+\tau) - x(t)| \geq a) \leq \frac{2R(0) - 2R(\tau)}{a^2}$ $\left(E[x(t+\tau)^2] \& E[x(t)^2] = R(0)\right)$ ✓

ii) $P(|x(t+\tau) - x(t)| \geq a) = 1 - \int_{-\infty}^{\infty} \int_{x_2-a}^{x_2+a} f(x_1, x_2; \tau)\, dx_1\, dx_2$ ✓

One minus the probability of x_1 being in $\pm a$ range of x_2

1. a) $R_{yy}(\tau) = R_{xx}(\tau) * h^*(-\tau) * h(\tau)$ $\quad \mathcal{L}[Y'(t)+2Y(t)] = \mathcal{L}[x(t)] \Rightarrow sY(s)+2Y(s) = X(s) \Rightarrow \frac{Y(s)}{X(s)} = \frac{1}{s+2}$

$h(t) = e^{-2t}u(t)$

$\therefore R_{yy}(\tau) = 5\,\delta(\tau) * h^*(-\tau) * h(\tau)$ (since x is WSS)

$= 5\,h^*(-\tau) * h(\tau) = 5\int_{-\infty}^{\infty} h^*(\tau-\beta)\,h(-\beta)\,d\beta = 5\int_{-\infty}^{\infty} e^{-2(\tau-\beta)}\,e^{2\beta}\,u(\tau-\beta)\,u(-\beta)$ (For $\tau<0$, integrate from $-\infty$ to τ; For $\tau\geq 0$, integrate from $-\infty$ to 0)

$= 5\int_{-\infty}^{\tau} e^{-2\tau+4\beta}\,d\beta = \frac{5}{4}e^{-2\tau+4\beta}\Big|_{-\infty}^{\tau}$

$= \frac{5}{4}e^{-2|\tau|}$ ✓

$R_{yy}(0) = E(Y(t)^2) = \frac{5}{4}$ ✓

$R_{xy}(\tau) = R_{xx}(\tau) * h^*(-\tau) = 5h^*(-\tau) = e^{2\tau}u(-\tau)$ (-2)

d) $E[x(t)^2] = 10 = \int_{-\infty}^{\infty} S_{xx}(\omega)\,d\omega$ $\quad H(\omega) = \frac{1}{(j\omega)^2+2j\omega+5} \Rightarrow |H(\omega)|^2 = \frac{1}{-\omega^2+2j\omega+5}\cdot\frac{1}{-\omega^2-2j\omega+5} = \frac{1}{(5-\omega^2)^2+4\omega^2}$

$S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega)$ $\quad = \frac{1}{\omega^4-6\omega^2+25}$

$\omega^4-6\omega^2+25 = (\omega^2-3)^2+16 \Rightarrow |H(\omega)|^2$ reaches its minimum at $\omega=\pm 3$ (-2)

Thus let x consentrate its power at $\omega=16 \Rightarrow S_{xx}(\omega) = 5(\delta(\omega-3)+\delta(\omega+3))$ (-2)

$\Rightarrow E(y(t)^2) = \frac{10}{16}$ //

Poisson_process & Markov chains

4 WLOG, Assume $t_3 \geq t_2 \geq t_1$

a) $E[\tilde{x}(t_1)\tilde{x}(t_2)\tilde{x}(t_3)] = E[(x(t_1)-\lambda t_1)(x(t_2-t_1)+x(t_1)-\lambda t_2)(x(t_3-t_2)+x(t_2-t_1)+x(t_1)-\lambda t_3)]$

$= E[\tilde{x}(t_1)(\tilde{x}(t_2-t_1)+\hat{x}(t_1))(\tilde{x}(t_3-t_2)+\tilde{x}(t_1)+\hat{x}(t_2-t_1)]$

$= E[(\tilde{x}(t_1)\tilde{x}(t_2-t_1)+\tilde{x}(t_1)^2)(\tilde{x}(t_3-t_2)+\tilde{x}(t_1-t_1)+\hat{x}(t_1))]$

$= E[\tilde{x}(t_1)\tilde{x}(t_2-t_1)\tilde{x}(t_3-t_2)+\tilde{x}(t_1)^2\tilde{x}(t_3-t_2)+\tilde{x}(t_1)\tilde{x}(t_2-t_1)^2+\tilde{x}(t_1)^2\hat{x}(t_2-t_1)+\tilde{x}(t_1)^3]$

All $\tilde{x}(t_i)$ terms with different t are independent by poisson process definition

$\therefore E[\tilde{x}(t_1)\tilde{x}(t_2)\tilde{x}(t_3)] = E[\tilde{x}(t_1)^3]$

$= E[x(t_1)^3 - 3\lambda t_1 x(t_1)^2 + 3(\lambda t)^2 x(t_1) - (\lambda t_1)^3]$

$= E(x(t_1)^3] - 3\lambda t_1(\lambda t_1+(\lambda t_1)^2) + 3(\lambda t_1)^3 - (\lambda t_1)^3$ since $E(x(t_1)^2) = \lambda t_1+(\lambda t_1)^2$ & $E(x(t_1)] = \lambda t_1$

$= E(x(t_1)^3] - 3(\lambda t_1)^2 - 3(\lambda t_1)^3 + 3(\lambda t_1)^3 - (\lambda t_1)^3$

$= E[x(t_1)^3] - 3(\lambda t_1)^2 - (\lambda t_1)^3$

$E(x(t_1)^3) = e^{-\lambda t_1}\sum_{n=0}^{\infty} n^3\frac{(\lambda t_1)^n}{n!} = e^{-\lambda t_1}\left(\sum_{n=1}^{\infty}\frac{n^2}{(n-1)!}(\lambda t_1)^n\right)$ ← limit changed as term is 0 when $n=1$

$= e^{-\lambda t_1}\left(\sum_{n=1}^{\infty}\frac{1}{(n-1)!}(\lambda t_1)^n + \sum_{n=2}^{\infty}\frac{n^2-1}{(n-1)!}(\lambda t_1)^n\right)$

$= e^{-\lambda t_1}\left(\lambda t_1 e^{\lambda t_1} + \sum_{n=2}^{\infty}\frac{n+1}{(n-2)!}(\lambda t_1)^n\right)$ ← definition of exponential

$= \lambda t_1 + e^{-\lambda t_1}\left(\sum_{n=2}^{\infty}\frac{n-2}{(n-2)!}(\lambda t_1)^n + \sum_{n=2}^{\infty}\frac{3}{(n-2)!}(\lambda t_1)^n\right)$

$= \lambda t_1 + e^{-\lambda t_1}\left((\lambda t_1)^3 e^{\lambda t_1} + 3(\lambda t_1)^2 e^{\lambda t_1}\right)$

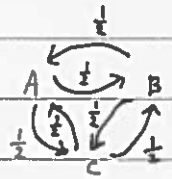
$= \lambda t_1 + (\lambda t_1)^3 + 3(\lambda t_1)^2$

$\therefore E(\tilde{x}(t_1)\tilde{x}(t_2)\tilde{x}(t_3)] = \lambda t_1$ if $t_1 \leq t_2 \leq t_3$

similarly $E(\tilde{x}(t_1)\tilde{x}(t_2)\tilde{x}(t_3) = \lambda t_2$ if t_2 is minimum

$= \lambda t_3$ if t_3 is minimum //

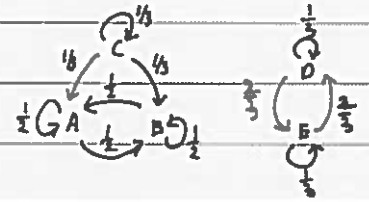
4. b) $P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$ ⇒



irreducible ? ✓1

States A, B, C are aperiodic recurrent positive states ✓

$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 2/3 & 1/3 & 0 \\ 1/3 & 1/3 & 0 & 0 & 1/3 \end{pmatrix}$



D, E is irreducible, aperiodic recurrent positive ✓

A, B is irreducible, aperiodic recurrent positive ✓

C is transient ✓

closed sets ? ✓1

Simulation of random walk for $p = \frac{1}{3}$

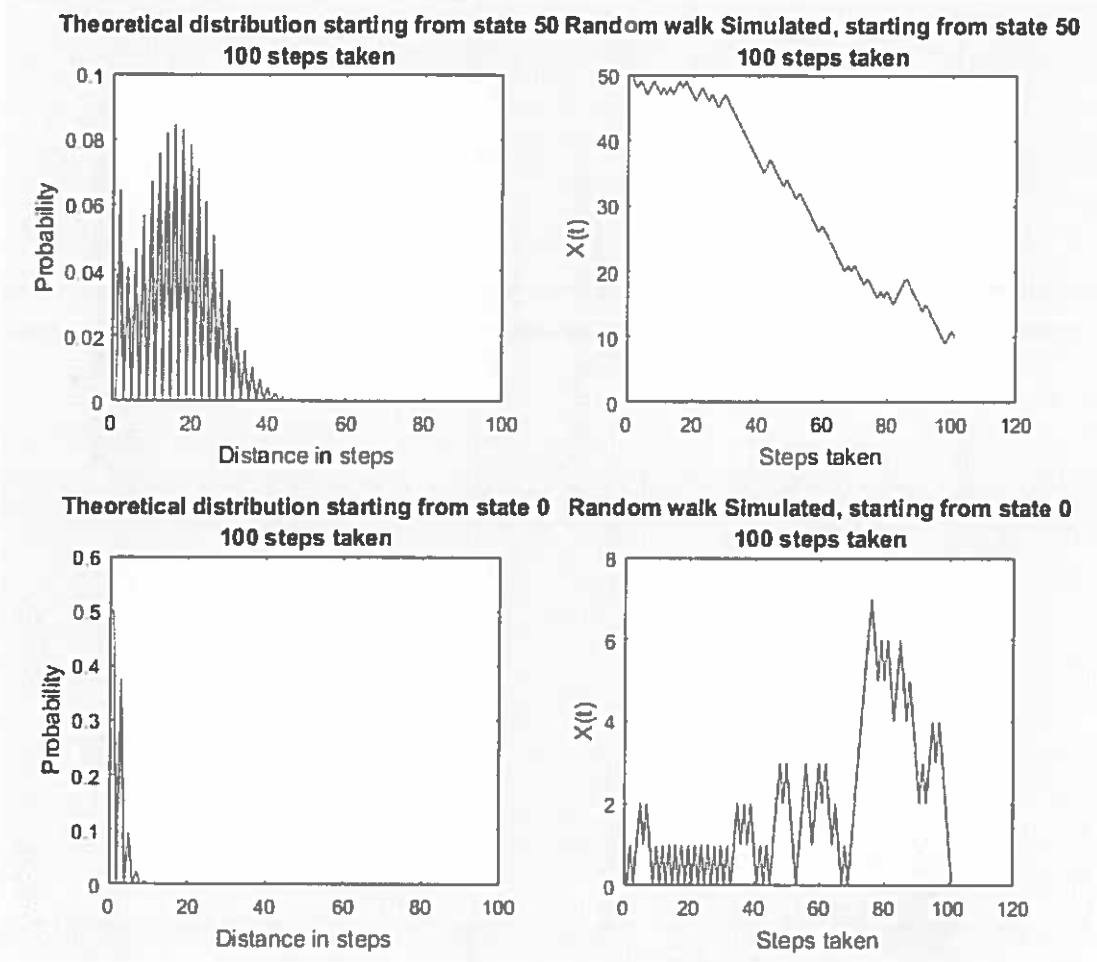


Figure 1, P=1/3, 100 Steps

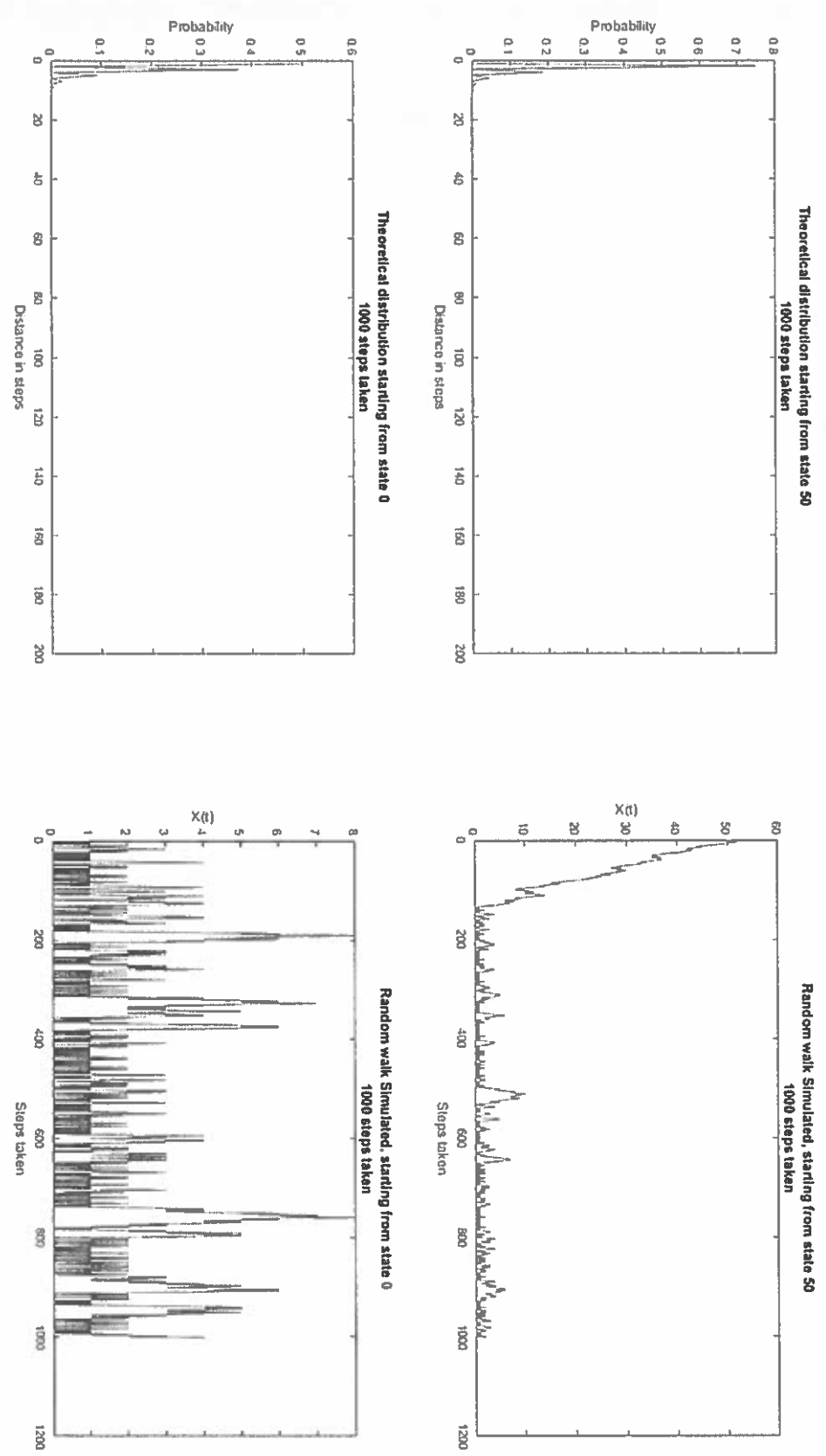


Figure 2,P=1/3, 1000 Steps

Simulation of random walk for $p = \frac{1}{2}$

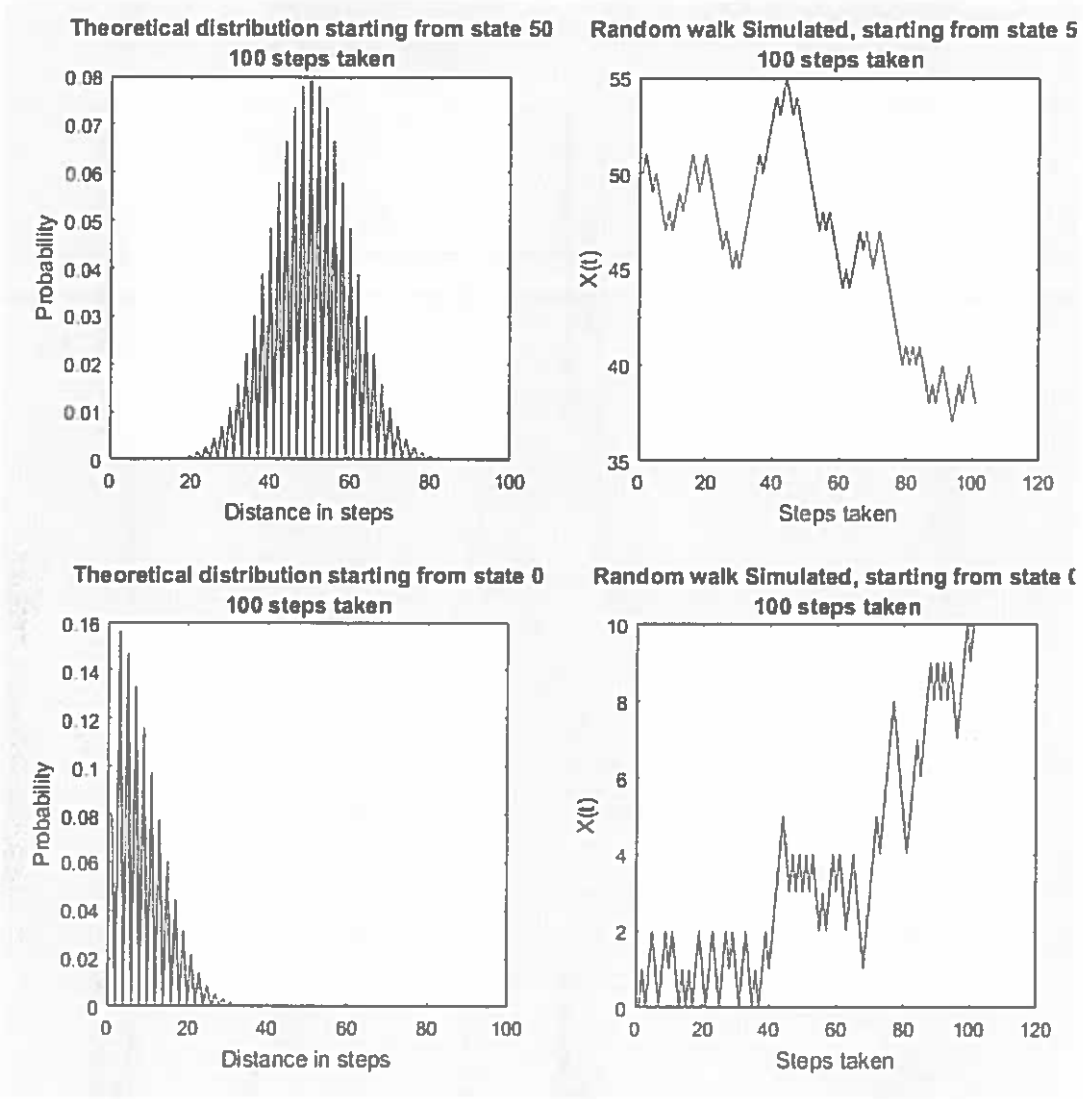


Figure 3, ,P=1/2, 100 steps

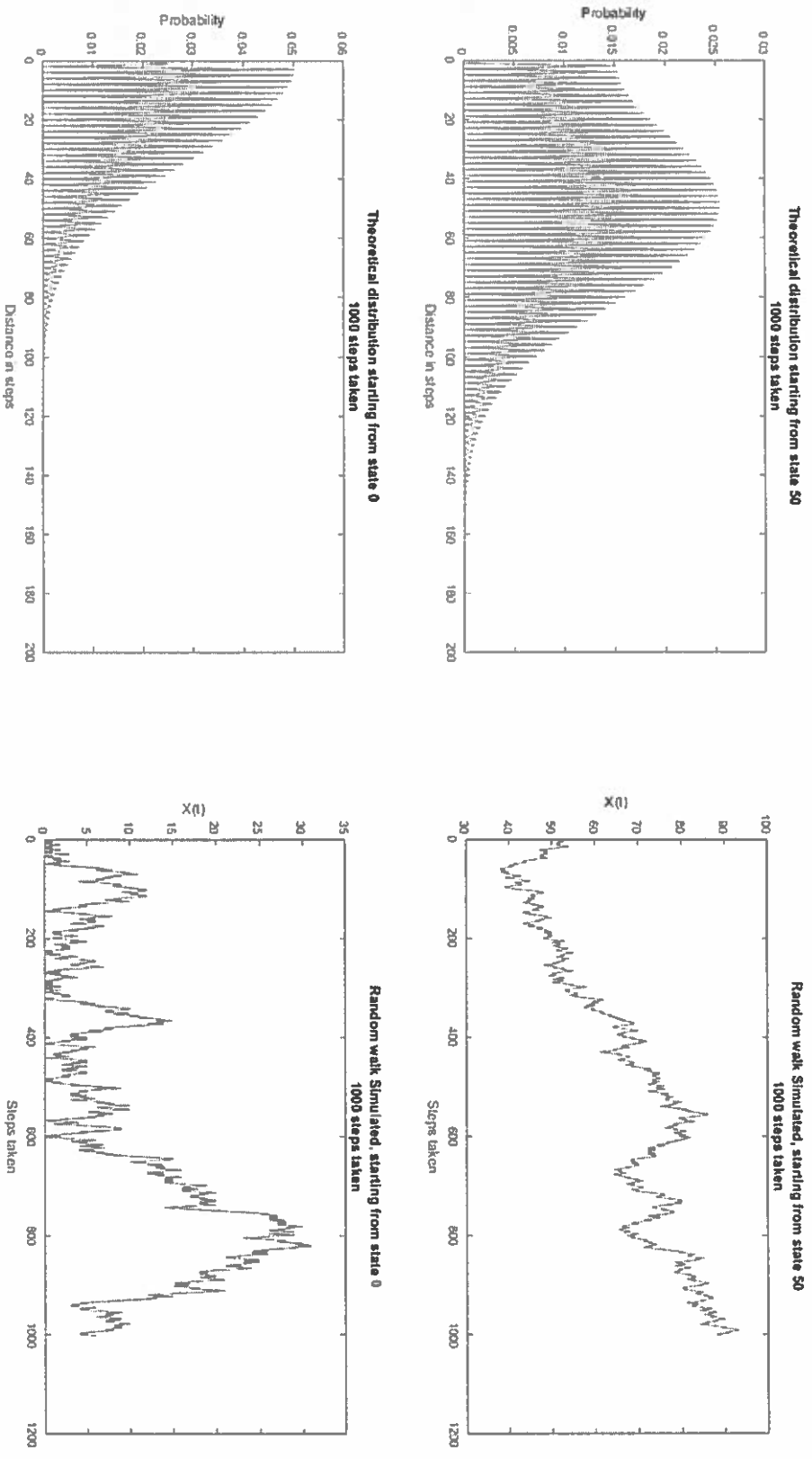


Figure 4, P=1/2, 1000 Steps

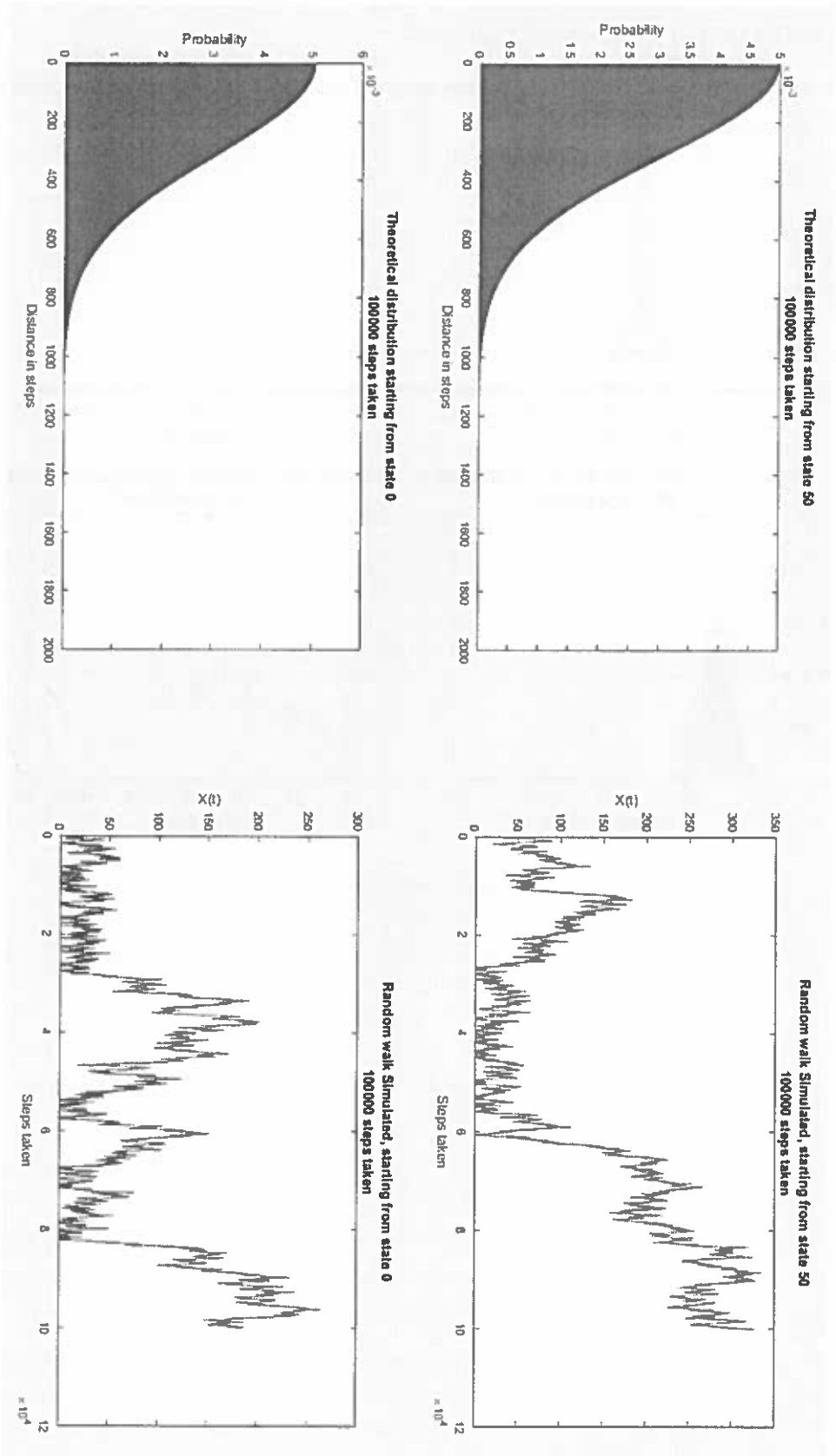


Figure 5, p=1/2, 100000 steps

Simulation of random walk for $p = \frac{2}{3}$

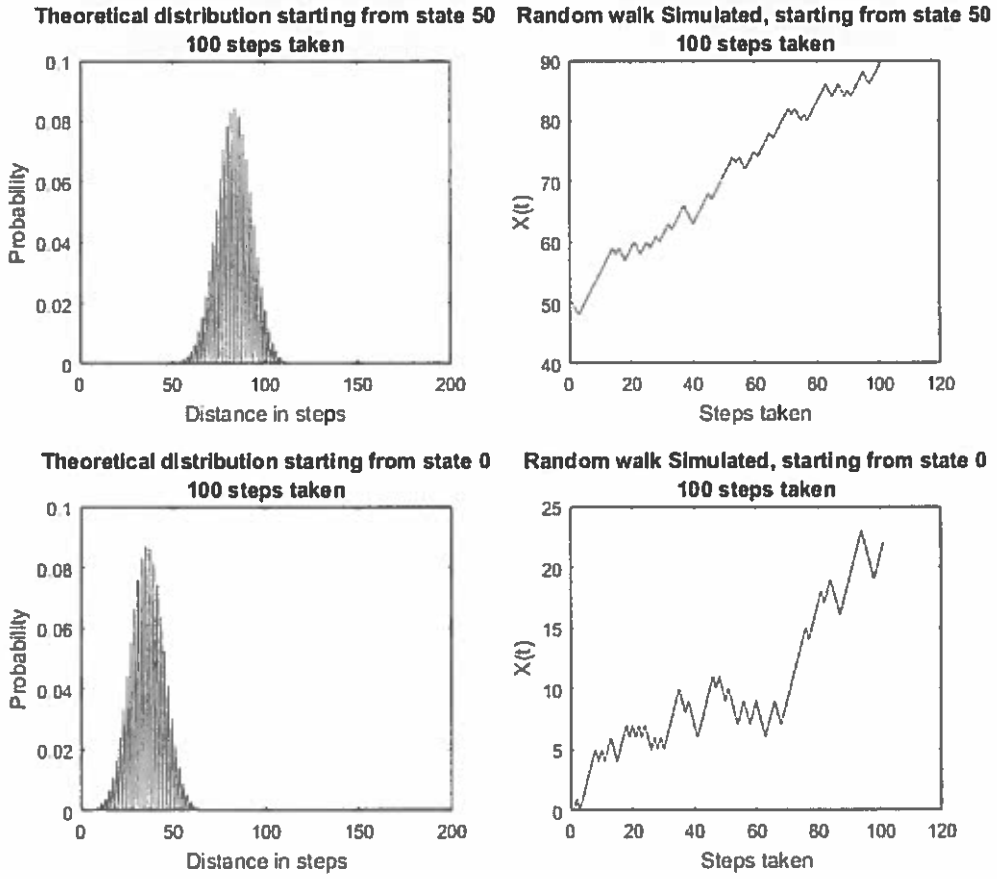


Figure 6,P=2/3, 100 steps

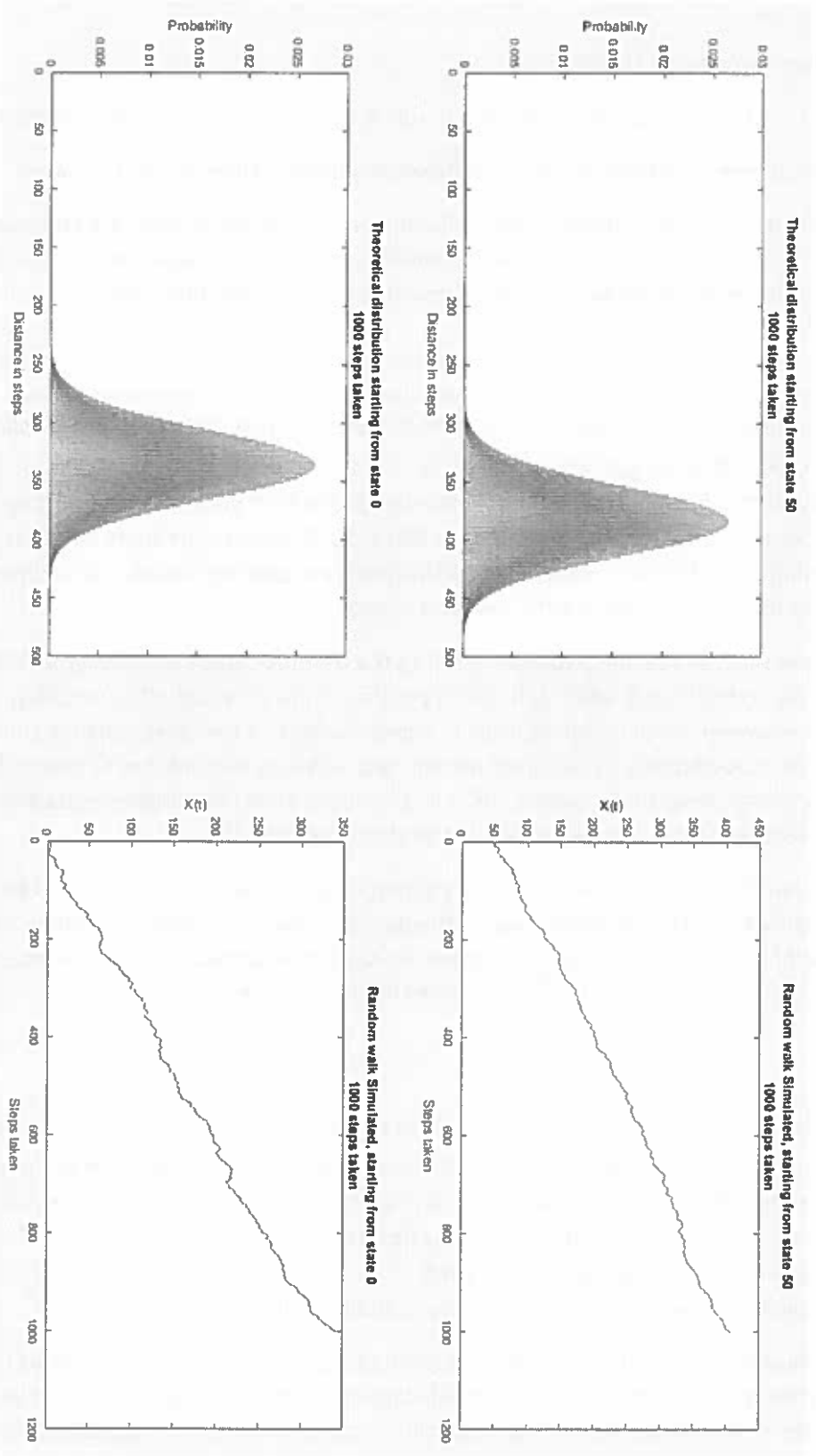


Figure 7,P=2/3, 1000 steps

Discussion

The random walk can be classified as:

$$x(t+1) = x(t) - 1, with\ probability\ (1-p)\ \&\ x(t+1) = x(t) + 1, with\ probability\ p$$

In this case, there is a barrier at the 0th state which always transits to the 1st state.

Due to the nature of the transition matrix, if odd or even number of steps are taken, it is only possible for the final state to be at odd or even distances in steps respectively. ($P_{ii} = 0$ implies that the state can never transition to itself resulting in the above statement.)

$p = \frac{1}{3}$

For $p = \frac{1}{3}$, since it is more likely for a -1 state transition compared to that of a $+1$ state transition, given that the starting position is far from the barrier position (0th state), it is expected that the final state will be closer (in steps) to the 0th state from the starting position. This can be seen in the random walk of both 100 & 1000 steps where there is a clear trend of transitioning closer to the 0th state in **Error! Reference source not found.** & **Error! Reference source not found.** where the starting position is state 50.

In the event that the starting position is close to the 0th state, there is a high probability of transitioning to the barrier which will tend transition to the 1st state with probability 1 on the next step. However, since it is more likely to transition back to the barrier than to the next higher state, this creates a situation where the state will tend to transition between the barrier and a few states closer to the barrier. This can be seen in **Error! Reference source not found.** & **Error! Reference source not found.** where the starting state is 0.

From the two figures, it can be seen that by taking a large number of steps, the probability distribution will tend to converge towards the barrier a few of its subsequent steps. This is indicative of a recurrent distribution. In this case, the barrier and its closest subsequent states are recurrent positive, however all other states are recurrent null.

$p = \frac{1}{2}$

For $p = \frac{1}{2}$, since it is equally likely for a -1 state transition and a $+1$ state transition to occur, given that the starting position is far from the barrier position (0th state), it is expected that the final state will be centred around the starting position. This can be seen in the random walk of both 100 steps where there is a clear trend of transitioning closer to the 0th state in Figure 3 where the starting position is state 50. The probability of ending the walk is the state far from the initial state in this case decreases geometrically[1].

In the case where the starting state is close (with reference to the number of steps) to the barrier, it is expected that there will be two local peaks at both the barrier and the original starting state. This can be seen in Figure 3 (starting position state 0) & Figure 4 (starting

[1] $P(|x(t) - starting| = n) \approx \sum_k \binom{k+n}{k} p^{k+n}(1-p)^k$, assuming starting state is far from barrier aka, the barrier does not exist

position state 0 & 50). The peak at the original starting state is explained in the above paragraph. The peak at the barrier is due to the probability distribution of states to the left of the barrier (negative states) being reflected around the barrier and added to the original distribution as such negative states cannot occur due to the barrier.

In the case where a large number of steps are taken and the starting state is far from the barrier, the ending states are all recurrent null the probability of each ending state decreases geometrically with each step. However, if the initial state is close to the barrier, the barrier and a few of its subsequent states will be recurrent positive while other states are recurrent null due to the back-and-forth state transitions near the barrier. This is seen in Figure 5, where 100000 steps are taken.

$$p = \frac{2}{3}$$

It is clear from the simulated random walk for $p = \frac{2}{3}$ that as it is more likely for a $+1$ state transition to occur compared to that of a -1 state transition. The final state of the random walk is highly likely to be at a much higher state than the initial state regardless of starting position. As such, if there is no bound on the number of steps taken, each subsequent state visit is likely to be different from the previously visited state. This would result in all states being transient. This is shown in Figure 6 & Figure 7 where the distribution of ending states is centred on different states with a different number of steps.

Conclusion

The results corroborate with the notes in which states close to the barrier are recurrent positive and all others recurrent null if $p < \frac{1}{2}$, recurrent positive for states near the barrier and null for all other states if $p = \frac{1}{2}$ & transient, including state 0 & 1, if $p > \frac{1}{2}$.